

Advanced Networking, Operating Systems & Cloud Fundamentals

13 Days Cyber Security Workshop • SJMIT College

SJM Institute of Technology
Department of Computer Science &
Engineering
Student Documentation

1. Advanced Networking Concepts

Subnetting vs. Supernetting

- **Subnetting:** The process of dividing a single large physical or logical IP network into multiple smaller, isolated subnetworks (subnets) to improve performance, routing efficiency, and security.
- **Supernetting:** The process of aggregating multiple smaller contiguous IP network blocks into a single larger routing prefix (CIDR aggregation) to reduce the size of router routing tables.

Routing & Routing Protocols

Routing is the systematic process of selecting paths across one or more networks along which to send network traffic.

Protocol	Full Form	Type	Description & Characteristics
RIP v1	Routing Information Protocol (v1)	Distance Vector (Classful)	Uses hop count as routing metric (maximum 15 hops). Does not support subnet masks.
RIP v2	Routing Information Protocol (v2)	Distance Vector (Classless)	Enhanced version supporting CIDR, variable-length subnet masks (VLSM), and authentication.
EIGRP	Enhanced Interior Gateway Routing Protocol	Advanced Distance Vector / Hybrid	Cisco proprietary protocol with fast convergence, low bandwidth overhead, and composite metrics.
IGRP	Interior Gateway Routing Protocol	Distance Vector (Classful)	Cisco legacy interior routing protocol (superseded by EIGRP).
BGP	Border Gateway Protocol	Path Vector	The core routing protocol of the global Internet, routing traffic between Autonomous Systems (AS).

2. Port Numbers & Categories

A **Port Number** is a 16-bit unsigned integer (ranging from 0 to 65535) used at the Transport Layer (TCP/UDP) to direct traffic to specific applications and services on a computer.

Port Range	Category Name	Purpose & Key Protocols
0 - 1023	Well-Known Ports	Reserved for core system services: HTTP (80), HTTPS (443), SSH (22), DNS (53), FTP (20/21), Telnet (23), SMTP (25).
1024 - 49151	Registered Ports	Assigned by IANA for specific vendor applications: MySQL (3306), RDP (3389), PostgreSQL (5432), WinRM (5985/5986).
49152 - 65535	Dynamic / Ephemeral Ports	Temporarily assigned by the client operating system for outbound client communication sessions.

3. Web State & Memory Concepts

- **Cookies:** Small key-value text files saved by web browsers on client devices to maintain user preferences, authentication tokens, and tracking data.
- **Sessions:** Server-side mechanisms that securely maintain user authentication state and interactions across multiple web requests.
- **Cache:** High-speed temporary storage that saves copies of frequently requested data to reduce retrieval latency and backend load.
- **Buffer:** A temporary region in physical RAM used to hold data while it is transferred between devices, applications, or network sockets.

4. Operating System Architecture & Management Modules

An **Operating System (OS)** manages computer hardware and provides common services for application programs.

#	OS Management Module	Core Function
1	Process Management	CPU scheduling, process creation, execution, synchronization, and termination.
2	Service Management	Managing background services, daemons (systemd, Windows services), and lifecycle.
3	Memory Management	Tracking memory allocation, virtual memory, paging, and swap space.
4	File System Management	Organizing files, directory trees, permissions, inodes, and disk partitions.
5	User & Account Management	User authentication, permissions, groups (UID/GID), and access controls.
6	Security Management	Antivirus integration, firewalls, permission enforcement, and security auditing.
7	Scripting & Automation	Executing shell scripts, batch commands, and cron/Task Scheduler jobs.
8	Network Management	Managing network interface cards (NICs), IP configurations, and socket connections.
9	Server Management	Resource balancing, multi-user concurrency, and high availability services.
10	Troubleshooting & Diagnostics	Event logging (Syslog, Event Viewer), core dump analysis, and performance monitoring.

Core OS Components & Boot Flows

- **Kernel:** Foundational engine of the OS interacting directly with the hardware (CPU, Memory, Disks).
- **Shell:** Command-line interface allowing users to issue instructions to the kernel.
- **Interpreter vs. Compiler:** Interpreters execute code line-by-line in real time; Compilers translate code into binary machine instructions beforehand.
- **Windows Boot Sequence:** BIOS/UEFI → MBR/GPT → Windows Boot Manager → winload.efi → NT Kernel (ntoskrnl.exe) → HAL → smss.exe → winlogon.exe.
- **Linux Boot Sequence:** BIOS/UEFI → GRUB Bootloader → Kernel Init → initramfs → systemd (PID 1) → Runlevel / Target.

5. Linux Distribution Families & Shells

Red Hat Family RHEL, CentOS, Fedora, Rocky Linux. Package Manager: RPM / DNF / YUM.	Debian Family Debian, Ubuntu, Kali Linux, Linux Mint. Package Manager: DPKG / APT.
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Comparison of Linux Shells

Shell	Full Name	Key Characteristics & Primary Use Case
bash	Bourne Again Shell	Standard default shell on most Linux systems; ideal for shell scripting.

sh	Bourne Shell	Original UNIX shell; standard for POSIX compatibility scripts.
zsh	Z Shell	Highly customizable interactive shell; popular with Oh-My-Zsh plugins and themes.
fish	Friendly Interactive Shell	User-friendly shell with auto-suggestions, web configuration, and syntax highlighting.
ksh	Korn Shell	Combines C-shell and Bourne shell features; common in enterprise UNIX environments.
tcsh / csh	C Shell / TENEX C Shell	Utilizes C-like programming syntax for scripting.
dash	Debian Almquist Shell	Lightweight, fast POSIX shell used for Debian/Ubuntu startup scripts.

6. Remote Operating System Connectivity Matrix

Client (Source)	Host (Destination)	Protocol	Port & Transport	Standard Tools & Utilities
Windows	Linux	SSH	22 / TCP	PuTTY, OpenSSH, Windows Terminal, PowerShell
Windows	Windows	RDP	3389 / TCP	Remote Desktop Connection (mstsc), PowerShell Remoting
Windows	Windows (Mgmt)	WinRM	5985 (HTTP) / 5986 (HTTPS)	PowerShell Remoting, Windows Admin Center
Windows	macOS	SSH / VNC	22 / 5900 TCP	PuTTY, VNC Viewer, Screen Sharing
macOS	Linux	SSH	22 / TCP	macOS Terminal, OpenSSH
macOS	Windows	RDP	3389 / TCP	Microsoft Remote Desktop for Mac
Linux	Linux	SSH	22 / TCP	Linux Terminal, OpenSSH
Linux	Windows	RDP	3389 / TCP	Remmina, FreeRDP, rdesktop

7. Cloud Computing Fundamentals & AWS Global Infrastructure

Cloud Computing is the on-demand delivery of IT resources (compute, database, storage, networking) over the Internet with pay-as-you-go pricing.

Cloud Service Provider Terminology Mapping

Domain	Amazon Web Services (AWS)	Microsoft Azure	Google Cloud Platform (GCP)
Compute / VMs	EC2 (Elastic Compute Cloud)	Azure Virtual Machines (VM)	Compute Engine
Object Storage	S3 (Simple Storage Service)	Azure Blob Storage	Cloud Storage
Resource Architecture	Services & Resource Groups	Resources & Resource Groups	Projects & APIs

AWS Global Infrastructure Architecture

Regions & Availability Zones (AZs)

- **Regions (26+ Worldwide):** Discrete geographic locations around the world where data center clusters reside.
- **Availability Zones (AZs):** Each Region has a minimum of **3** (and up to **6**) Availability Zones. Each AZ consists of one or more physical data centers equipped with independent power, cooling, and physical security, connected via ultra-low-latency redundant networking.

Criteria for Choosing a Cloud Region

1. **Latency & Proximity:** Minimizing distance to end-users to reduce response time.
2. **Service Availability:** Ensuring specific cloud services and instance types are supported in that region.
3. **Compliance & Data Sovereignty:** Meeting regulatory and legal requirements (e.g., GDPR, HIPAA, local data privacy acts).
4. **Cost & Pricing:** Varying operational and tax rates across different geographic locations.